Florian (Rian) M Browne III

Ph.D. Candidate, School of Mechanical Engineering, Purdue University

Contact Information

School of Mechanical Engineering
Purdue University
Herrick Laboratories
177 S Russell St,
West Lafayette, IN 47907

Phone: 785-554-3588
E-mail: fbrowne@purdue.edu

Education

May 2020	**Ph.D.** in **Mechanical Engineering** Purdue University, West Lafayette, IN *GPA:* 3.80/4.00 *Advisor:* Prof. Neera Jain *Dissertation:* "Robust Iterative Learning Control for Linear Parameter-Varying Systems with Time Delays"
May 2015	**B.S.** in **Mechanical Engineering** Kansas State University, Manhattan, KS *Honors:* Summa Cum Laude, *GPA:* 3.966/4.000

Work Experience

Research Assistant, Purdue University, West Lafayette, IN, Aug. 2015-Present

- Created and validated a reduced order model of the twin roll strip casting process, also known as the Castrip process, using MATLAB
- Developed a wedge controller using an iterative learning control (ILC) algorithm that is robust to measurement delay estimation error
- Implemented and tested the wedge controller on a twin roll casting process at Nucors Castrip plant in Crawfordsville, IN, using LabVIEW
- Reduced the measured wedge by 50% compared to Nucor's previous wedge controller
- Developed a fault detection and response algorithm that supervises the wedge controller and enables the deployment of the wedge controller to more plants using the Castrip process.

Control Theory Co-op, Fisher Controls International LLC, Marshalltown, IA, Jan. 2014-Aug. 2014

- Used my understanding of feedback control theory, control valves, actuators, compressible flow, firmware, and software to assist in the development of a new positioner (servocontroller)
- Assisted in the development and debugging of the calibration procedures for the new positioner
- Established the tuning sets for the new servocontroller on various actuator sizes
- Collected raw data and performed data reduction, using R, to present my results to my project group
- Created test procedures and acceptance criteria for the new servocontroller

- Helped market the new positioner to Chevron and ExxonMobil
- Programmed a CVI program to test valve-actuator assemblies
- Rewired DAQ boxes to create a safer work space

Research Interests

- Iterative learning control (ILC)
- Time-delay systems
- Manufacturing systems

Skills

- Experience in many coding languages: R, C, C++, MATLAB, CVI, LabVIEW, Simulink, Python
- Proficient in Microsoft Office, including Visio
- Certified in Solid Works
- Experience with microcontrollers, including soldering experience
- Extensive experience in presenting to both large and small groups

Awards and Recognition

- **Laura Winkelman Davidson Fellowship**, Purdue University, 2015. Awarded to an exceptional graduate student in the School of Mechanical Engineering.

Publications

Peer Reviewed Journal Articles

J1 **F. Browne**, G.T.-C. Chiu, and N. Jain, "A Nonlinear Dynamic Switched-Mode Model of Twin-Roll Steel Strip Casting." *ASME Journal of Dynamic Systems, Measurement and Control*, 2019. doi:10.1115/1.4042952.

J2 **F. Browne**, G.T.-C. Chiu, and N. Jain, "Cognitive Modeling of Human Trust in Human-Machine Interactions." *IEEE Transactions on Control Systems Technology*(Under review).

Peer Reviewed Conference Papers

C1 **F. Browne**, G.T.-C. Chiu, and N. Jain, "Dynamic Modeling of Twin-roll Steel Strip Casting." Proceedings of the 2016 ASME Dynamic Systems and Control Conference, Minneapolis, MN, October 12-14, 2016.

C2 **F. Browne**, G.T.-C. Chiu, and N. Jain, "Iterative Learning Control For Periodic Disturbances in Twin-Roll Strip Casting with Measurement Delay." Proceedings of the 2018 American Control Conference, Milwaukee, WI, June 27-29, 2018.

C3 **F. Browne**, B. Rees, G.T.-C. Chiu, and N. Jain, "ILC With Time-Varying Delay Estimation: A Case Study on Twin Roll Strip Casting." Proceedings of the 2018 ASME Dynamic Systems and Control Conference, Atlanta, GA, September 30 - October 3, 2018.

C4 **F. Browne**, G.T.-C. Chiu, and N. Jain, “A Coupled Adaptive Measurement Delay Estimation and Iterative Learning Control Algorithm.” Proceddings of the 2019 ASME Dynamic Systems and Control Conference, Park City, UT, October 8-11, 2019.

Patents

P1 **F.M. Browne III**, G.T.C. Chiu, N. Jain, H.B. Rees, “Iterative Learning Control for Periodic Disturbances in Twin-Roll Strip Casting With Measurement Delay.” U.S. Patent US20190091761A1.

Service at Purdue University

Mentorship

- 2017: Purdue Graduate School’s eMentoring Program
- 2018 – Present: Official Mechanical Engineering Graduate Students Association (OMEGA)

Select Volunteer Work

- **Coder Dojo** June 2019
  - Assist in teaching children between the ages 7-17 how to write computer scripts and mechatronics
- **Welcome to Scouting Event** September 2018
  - Help new cub scouts assemble and launch electric rockets

Professional Activities (External Service)

Conference Reviewer

- 2018 – present: American Control Conference (ACC)
- 2016 – present: ASME Dynamic Systems and Control Conference (DSCC)

Journal Reviewer

- 2019 – present: Journal of Dynamic Systems, Measurement, and Control (JDSMC)

Professional Society Memberships

- American Society of Mechanical Engineers (ASME), Dynamic Systems and Control Division
- Tau Beta Pi National Engineering Honor Society (TBP)
- Pi Tau Sigma Mechanical Engineering Honor Society (PTS)